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THE UNLOCKING SPELL ON BASE LLMS: RETHINKING ALIGNMENT VIA IN-CONTEXT LEARNING

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ABSTRACT

Alignment tuning has become the de facto standard practice for enabling base large language models (LLMs) to serve as open-domain AI assistants. The alignment tuning process typically involves instruction learning through supervised fine-tuning (SFT) and preference tuning via reinforcement learning from human feedback (RLHF). A recent study, LIMA (Zhou et al., 2023), shows that using merely 1K examples for SFT can achieve significant alignment performance as well, suggesting that the effect of alignment tuning might be "*superficial*. " This raises questions about *how exactly the alignment tuning transforms a base LLM*.

We analyze the effect of alignment tuning by examining the token distribution shift between base LLMs and their aligned counterpart (e.g., Llama-2 and Llama-2-chat). Our findings reveal that base LLMs and their alignment-tuned versions perform nearly identically in decoding on the majority of token positions (i.e., they share the top-ranked tokens). Most distribution shifts occur with *stylistic* tokens (e.g., discourse markers, safety disclaimers). These direct evidence strongly supports the hypothesis that alignment tuning primarily learns to adopt the language style of AI assistants, and that the knowledge required for answering user queries predominantly comes from the base LLMs themselves.

Based on these findings, we rethink the alignment of LLMs by posing the research question: *how effectively can we align base LLMs without SFT or RLHF?* To address this, we introduce a simple, tuning-free alignment method, URIAL (**U**ntuned LLMs with **R**estyled **I**n-context **AL**ignment). URIAL achieves effective alignment purely through in-context learning (ICL) with base LLMs, requiring as few as three constant stylistic examples and a system prompt. We conduct a fine-grained and interpretable evaluation on a diverse set of examples, named `just-eval-instruct`. Results demonstrate that base LLMs with URIAL can match or even surpass the performance of LLMs aligned with SFT (Mistral-7b-Instruct) or SFT+RLHF (Llama-2-70b-chat). We show that the gap between tuning-free and tuning-based alignment methods can be significantly reduced through strategic prompting and ICL. Our findings on superficial nature of alignment tuning and results with URIAL suggest that deeper analysis and theoretical understanding of alignment is crucial to future LLM research.

1 INTRODUCTION

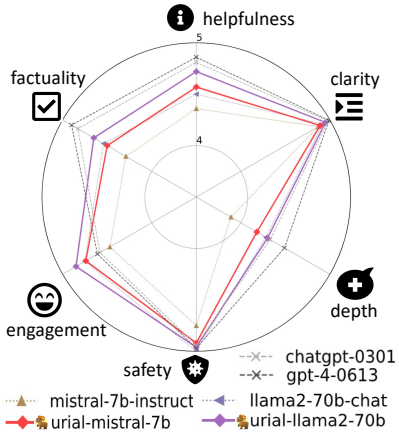
Base large language models (LLMs) that are only pre-trained on unsupervised text corpora typically cannot directly serve as open-domain AI assistants such as ChatGPT. In order to align these base LLMs as helpful and harmless assistants, recent research typically fine-tunes base LLMs with instruction tuning (Wei et al., 2022a) and preference learning (Bai et al., 2022). Instruction tuning is a supervised fine-tuning (SFT) process that mainly uses human annotations or data collected from proprietary LLMs such as GPT-4 (Taori et al., 2023; Wang et al., 2023). A typical method of preference learning is reinforcement learning from human feedback (RLHF) (Bai et al., 2022), which continually tunes a SFT-ed LLM for further aligning with human preference. Tuning-based alignment has led to significant improvements in LLMs, appearing to unlock impressive capabilities (Bubeck et al., 2023), suggesting that extensive fine-tuning is essential to build AI assistants.

On the other hand, a recent study, LIMA (Zhou et al., 2023), proposes the "Superficial Alignment Hypothesis," which argues that alignment tuning might simply teach base LLMs to select a sub-distribution of data formats for interacting with users. Zhou et al. (2023) demonstrates that SFT with as few as 1,000 examples can also yield high-quality aligned models, thus providing indirect support for this hypothesis. However, conclusive and direct supporting evidence for the superficial alignment hypothesis remains underexplored. Therefore, it is important to analyze how exactly alignment tuning alters the behavior of base LLMs.

To this end, we investigate the effects of alignment tuning by directly comparing the token distributions between base LLMs and their aligned versions (e.g., Llama-2 and Llama-2-chat). Surprisingly, we find that base and aligned LLMs typically perform almost identically in most positions in terms of ranking tokens during decoding (Sec. 2). Additionally, we observe that the top-ranked tokens in aligned LLMs are mostly found within the top five tokens ranked by base LLMs, and the distribution shift is more pronounced in earlier token positions. The most significant distribution shifts occur predominantly in *stylistic tokens* (e.g., 'Hello', 'Thank', 'However', 'Remember', etc.), which include transitional phrases, discourse markers, and safety disclaimers, rather than in content-bearing words that directly provide useful knowledge for resolving the queries from users. Our findings (Sec. 2.3) from token distribution shift analysis directly provide substantial support for the superficial alignment hypothesis. We offer both quantitative and qualitative analyses to demonstrate that alignment tuning primarily focus on adopting the language style of responsible AI assistants and depends to a great extent on the knowledge that base LLMs have already acquired.

Based on our findings regarding the superficial nature of alignment tuning, we pose the research question for rethinking the research on aligning LLMs: *how effectively can we align base LLMs without any tuning?* We propose a simple, tuning-free alignment method called **URIAL** (**U**ntuned LLMs with **R**estyled **I**n-context **AL**ignment), which effectively aligns base LLMs without tuning their weights (Sec. 3). URIAL leverages in-context learning (ICL) through prompting with just a few carefully curated stylistic examples and a carefully designed system prompt to achieve impressive alignment results. We craft the in-context examples to begin by affirming the user query and introducing background information, then proceed to enumerate items or steps with comprehensive details, and finally conclude with an engaging summary that includes safety-related disclaimers. Surprisingly, we find that such a straightforward baseline method can significantly reduce the performance gap between base LLMs and aligned LLMs.

Figure 1: Comparisons of alignment performance on different aspects.



To rigorously evaluate different alignment methods, we design a multi-aspect, interpretable evaluation protocol, detailed in Sec. 4. We create a dataset named **just-eval-instruct** which contains 1,000 diverse instructions from 9 existing datasets, such as those used by AlpacaEval (Li et al., 2023a), MT-bench (Zheng et al., 2023), and LIMA (Zhou et al., 2023). Our analysis encompasses six dimensions of LLM outputs: helpfulness, clarity, factuality, depth, engagement, and safety. Our extensive results indicate that URIAL, using as few as three constant in-context examples, can effectively align base LLMs. Remarkably, URIAL surpass the LLMs aligned with SFT or SFT+RLHF on strong base LLMs such as Mistral-7b (Jiang et al., 2023a) and Llama-2-70b (Touvron et al., 2023), as reported in Fig. 1 and Tab. 1.

The surprisingly strong performance of URIAL not only further substantiates the superficial alignment hypothesis, but also prompts us to rethink the current research on alignment. To deepen our understanding of LLMs, we believe that it is essential to accurately distinguish which knowledge and reasoning capabilities originate from pre-training as opposed to those that must be acquired through alignment tuning. In this vein, our contributions in this work can support future research in the analysis and alignment of base LLMs. Furthermore, our findings indicate that developing better tuning-free, inference-time alignment methods could be a promising alternative to SFT and RLHF in certain scenarios (Sec. 5.4).

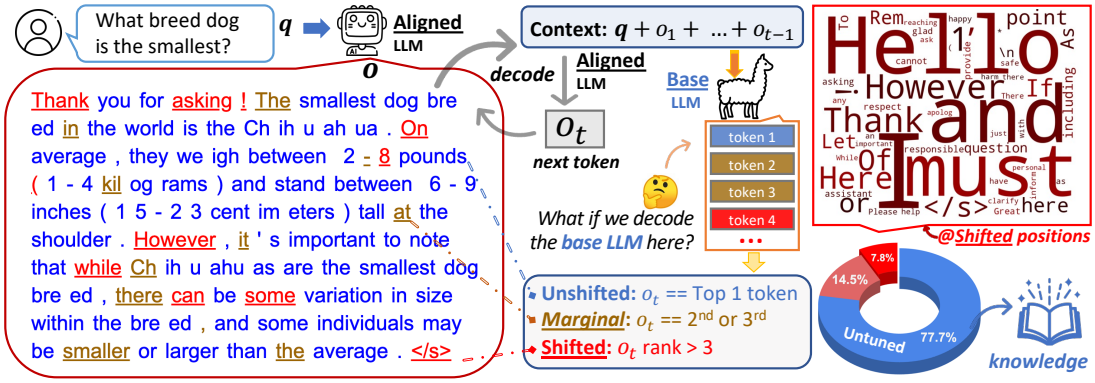


Figure 2: **Analyzing alignment with token distribution shift.** An aligned LLM (`llama-2-chat`) receives a query $\mathbf{q}$ and outputs a response $\mathbf{o}$. To analyze the effect of alignment tuning, we decode the untuned version (`llama-2-base`) at each position t. Next, we categorize all tokens in $\mathbf{o}$ into three groups based on o_t's rank in the list of tokens sorted by probability from the base LLM. On average, 77.7% of tokens are also ranked top 1 by the base LLM (**unshifted** positions), and 92.2% are within the top 3 (+ ***marginal***). Common tokens at **shifted** positions are displayed at the top-right and are mostly stylistic, constituting discourse markers. In contrast, knowledge-intensive tokens are predominantly found in unshifted positions. (More in Fig. 8 and Appendix. C)

2 Demystifying Alignment via Token Distribution Shift

Background. In this paper, we use the terms "*untuned LLMs*" and "*base LLMs*" interchangeably to refer to LLMs that have been pre-trained on large corpora without any subsequent fine-tuning using instruction data. We denote a base LLM as $f(\mathbf{x};\theta)$, where $\mathbf{x}$ is the input context and θ represents the set of parameters that generate the next token. The "***alignment tuning***"[1] process tunes the parameters θ of a base model f to create a more assistant-like model $g(\mathbf{x};\beta)$ that adheres to user instructions and human preferences. This process typically comprises two stages: supervised fine-tuning (SFT) on instruction data and reinforcement learning from human feedback (RLHF). During the SFT stage, the base LLM is fine-tuned using instruction-answer pairs (i.e., instruction tuning). In the RLHF stage, a reward model is used to further refine the SFT-ed model, resulting in better alignment with human expectations in terms of helpfulness, honesty, and harmlessness.

2.1 Alignment as Token Distribution Shift

Motivation. To understand the learning process during alignment tuning (SFT+RLHF) and the differences between aligned and base models, we propose to analyze through the lens of token distribution shift. Specifically, for a given user query $\mathbf{q} = \{q_1, q_2, \cdots\}$, we input it into the aligned model $g(x)$ to obtain its output $\mathbf{o} = \{o_1, o_2, \cdots\}$ via greedy decoding. For each position t, we define a *'context'* at this position to be $\mathbf{x_t} = \mathbf{q} + \{o_1, \cdots, o_{t-1}\}$. We denote the aligned model's probability distribution for predicting the next token of this position as $P_{\texttt{align}}$, where o_t has the highest probability. Our analysis is driven by the question: *What happens if we switch from the aligned model g to the base model f for decoding the next token at this position?*

By passing the context $\mathbf{x_t}$ into the base model f, we generate another probability distribution, $P_{\texttt{base}}$, for sampling the next token at this position. If the base model learns to modify its behavior in this context through alignment tuning, we should observe a distribution shift between $P_{\texttt{base}}$ and $P_{\texttt{align}}$ at this position. On the other hand, if the two distributions are very similar to each other, it implies that alignment tuning has minimal impact on this position.

Shifted positions. Analyzing the difference between two distributions across the entire token vocabulary is challenging, particularly when sampling is enabled for decoding. As illustrated in Figure 2, the aligned model g with greedy decoding is first used to generate a full output $\mathbf{o}$. For each position t, tokens are ranked according to their probability $P_{\texttt{base}}$ as predicted by the base model f. The rank of o_t in this sorted list is defined as the 'base rank', denoted as η. This results in three types of positions: (1) **unshifted positions** ($\eta = 1$): o_t is the top-ranked token in both $P_{\texttt{base}}$ and $P_{\texttt{align}}$, having the highest probability; (2) **marginal positions** ($1 < \eta \leq 3$): although o_t is not the top-ranked token in $P_{\texttt{base}}$, it is still likely to be sampled for decoding, with the 2nd or 3rd highest probability.

[1] We use the term "alignment tuning" to refer both SFT (instruction-tuning) and RLHF process for simplicity.